NAAN MUDHALVAN - SALESFORCE DEVELOPER

PROJECT DOCUMENTATION

FARM MANAGEMENT SYSTEM

A Salesforce Lightning Application for Managing Farms, Crops, Farmers and Buyers

PROJECT DETAILS

Project Title	**FARM MANAGEMENT SYSTEM**
Submitted by	Nandhini.M (Team Member)
Department	III Year B.Sc. Computer Science
Semester	5
College	Immaculate College for Women, Cuddalore
University	Annamalai University
College Code	ANM121

TEAM DETAILS

S.No	Name	Register No	Naan Mudhalvan ID	Role
1	**Sriraksha L.M.**	24131211802522079	220F8C6F860B21D9E7545692480A6C1C	Team Leader
2	**Nandhini M**	24131211802522045	4208536952000B89F5C419A746CB7BFD	Member
3	**Sivapriya S**	24131211802522076	49AC9F5B99B11F3A67043DD30385C0D9	Member
4	**Usharani I**	24131211802522083	054AE04EABAB59FA7FC41847F55EFC04	Member

IMMACULATE COLLEGE FOR WOMEN

Academic Year 2026 - 2027

ABSTRACT

The Farm Management System is a custom Lightning application developed on the Salesforce platform. It helps a farm manager keep the records of farms, crops, farmers and buyers in one secure place and view simple summaries of them.

The system uses four custom objects: Farm, Crop, Farmer and Buyer. A Master-Detail relationship connects each farm to its farmer, and a Lookup relationship connects each crop to its farm. A dependent picklist makes the District choices depend on the State, and a formula field calculates the expected yield per acre automatically.

Access to data is controlled by the Farm Manager profile and role together with sharing rules for the Crop and Buyer objects. The Farm Management System app brings six navigation items together: Farms, Crops, Farmers, Buyers, Reports and Dashboards. The Farmer With Farms report and dashboard summarise which farms belong to which farmers.

The whole solution was configured without writing code, using Object Manager, App Manager, Report Builder and Dashboard Builder in a Salesforce Developer Edition organisation. The project shows how a cloud platform can replace scattered paper and spreadsheet records with a connected and secure system.

Keywords: *Salesforce, Lightning App, Custom Objects, Master-Detail, Lookup, Dependent Picklist, Sharing Rules, Report, Dashboard*

TABLE OF CONTENTS

CHAPTER 1: INTRODUCTION 6
1.1 Overview 6
1.2 About the Salesforce Platform 6
1.3 Problem Statement 6
1.4 Objectives 7
1.5 Scope of the Project 7
1.6 Organisation of the Report 7
CHAPTER 2: SYSTEM ANALYSIS 9
2.1 Existing System 9
2.2 Limitations of the Existing System 9
2.3 Proposed System 9
2.4 Advantages of the Proposed System 9
2.5 Feasibility Study 10
2.6 Software Requirements 10
2.7 Hardware Requirements 10
2.8 Functional Requirements 10
CHAPTER 3: SYSTEM DESIGN 12
3.1 System Architecture 12
3.2 Data Model 12
3.3 Object Design 13
3.4 Dependent Picklist Design 14
3.5 Security Design 14
3.6 User Interface Design 14
3.7 Report and Dashboard Design 14
CHAPTER 4: IMPLEMENTATION 15
4.1 Implementation Overview 15
4.2 Farm Object - Fields and Relationships 16
4.3 Crop Object - Fields and Relationships 17
4.4 Farmer Object - Fields and Relationships 18
4.5 Buyer Object - Fields and Relationships 19
4.6 Dependent Picklist - State and District 20
4.7 Custom Object Tabs 21
4.8 Lightning Component Tabs 22
4.9 Lightning Experience App Manager 23
4.10 App Details and Branding 24
4.11 App Navigation Items 25
4.12 Profiles 26
4.13 Roles and Role Hierarchy 27
4.14 Users 28
4.15 Sharing Rule for the Crop Object 29
4.16 Sharing Rule for the Buyer Object 30
4.17 Farms List View 31
4.18 Farm Record - Green Farm 32
4.19 Crop Record - Rice 33
4.20 Farmer Record - Durgaseetha 34
4.21 Buyer Record - LMS Traders 35
4.22 Report - Farmer With Farms 36

4.23 Dashboard - Farmers With Farms 37
4.24 App Launcher 38
CHAPTER 5: TESTING 39
5.1 Testing Objectives 39
5.2 Test Environment 39
5.3 Functional Test Cases 39
5.4 Security Configuration Checks 40
5.5 Test Summary 40
ADVANTAGES AND DISADVANTAGES 41
CHAPTER 6: RESULTS AND DISCUSSION 42
6.1 Results Achieved 42
6.2 Discussion 42
6.3 Limitations 42
CHAPTER 7: CONCLUSION AND FUTURE ENHANCEMENTS 43
7.1 Conclusion 43
7.2 Future Enhancements 43
7.3 Learning Outcomes 43
APPENDIX 44
REFERENCES 45

LIST OF FIGURES

Figure 3.1: Entity-relationship diagram of the Farm Management System 12
Figure 4.2: Farm object: Fields & Relationships page 16
Figure 4.3: Crop object: Fields & Relationships page..... 17
Figure 4.4: Farmer object: Fields & Relationships page..... 18
Figure 4.5: Buyer object: Fields & Relationships page..... 19
Figure 4.6: Farmer Field Dependencies page..... 20
Figure 4.7: Setup > Tabs: custom object tabs 21
Figure 4.8: Setup > Tabs: Lightning component and page tabs..... 22
Figure 4.9: Lightning Experience App Manager 23
Figure 4.10: Lightning App Builder: App Details & Branding..... 24
Figure 4.11: Lightning App Builder: Navigation Items..... 25
Figure 4.12: Setup > Profiles: Farm Manager profile 26
Figure 4.13: Setup > Roles: role hierarchy 27
Figure 4.14: Setup > Users: All Users list..... 28
Figure 4.15: Setup > Sharing Settings: Crop Sharing Rules..... 29
Figure 4.16: Setup > Sharing Settings: Buyer Sharing Rules 30
Figure 4.17: Farm Management System app: Farms list view 31
Figure 4.18: Farm record: Green Farm 32
Figure 4.19: Crop record: Rice 33
Figure 4.20: Farmer record: Durgaseetha 34
Figure 4.21: Buyer record: LMS Traders..... 35
Figure 4.22: Report: Farmer With Farms..... 36
Figure 4.23: Dashboard: Farmers With Farms..... 37
Figure 4.24: App Launcher showing Farm Management System..... 38

CHAPTER 1: INTRODUCTION

1.1 Overview

The Farm Management System is a Salesforce-based application that brings the day-to-day records of a farming business into one unified digital record. It manages four kinds of information: farms, crops, farmers and buyers. Every record is linked to the others, so a user can start from a farmer and reach his farms, or start from a crop and find the farm and the farmer behind it.

The application was built entirely with the declarative tools of the Salesforce platform. Custom objects, fields, relationships, a dependent picklist, a formula field, a profile, a role, sharing rules, a Lightning app, a report and a dashboard were configured from the Setup menu without writing any program code.

The project was developed by a team of four students as part of the Naan Mudhalvan Salesforce Developer programme, using a Salesforce Developer Edition organisation.

1.2 About the Salesforce Platform

Salesforce is a cloud-based platform that is best known for customer relationship management. Beyond CRM, it offers a platform-as-a-service layer that lets a developer or administrator build custom business applications that run on the same secure cloud. Because the platform is multi-tenant and hosted by Salesforce, there is no server to install or maintain.

The platform has several features that made it suitable for this project:

- **Object Manager:** lets a user define custom objects and fields, which work like tables and columns in a database.
- **Relationships:** Lookup and Master-Detail relationships connect objects to each other.
- **Lightning App Builder:** groups tabs into a branded application with its own navigation bar.
- **Security model:** profiles, roles and sharing rules decide who can see and edit each record.
- **Reports and dashboards:** turn the stored records into summaries and visual components.

1.3 Problem Statement

Small and medium farms usually keep their records in notebooks or spreadsheets. As the number of farms and crops grows, this approach creates the following problems:

- Records are scattered, and the same farmer or farm is written down in many places.
- There is no reliable link between a crop, the farm where it was grown and the farmer who owns that farm.

- Anyone who has the file can see everything; there is no control over who views or edits a record.
- Yield figures such as the expected yield per acre must be calculated by hand every time.
- Producing a simple summary, for example which farmer owns which farm, takes a lot of manual work.

A structured, secure and easily searchable system is therefore needed.

1.4 Objectives

The main objective of the project is to design and build a working farm management application on the Salesforce platform. The specific objectives are:

- To store farm, crop, farmer and buyer details in separate, well-defined custom objects.
- To connect the objects using a Master-Detail relationship (Farm to Farmer) and a Lookup relationship (Crop to Farm).
- To reduce data-entry errors by using a dependent picklist in which the District list depends on the State chosen.
- To calculate the expected yield per acre automatically with a formula field.
- To control access using a custom profile, a role in the role hierarchy and sharing rules.
- To provide one Lightning app with six navigation tabs: Farms, Crops, Farmers, Buyers, Reports and Dashboards.
- To present the data through a report and a dashboard.

1.5 Scope of the Project

Within the scope

- Four custom objects with their fields and relationships.
- Field-level automation using a formula field and a field dependency.
- Role-based and rule-based record sharing.
- A Lightning app, a report (Farmer With Farms) and a dashboard (Farmers With Farms).

Outside the scope

- Custom programming using Apex or Lightning Web Components.
- Online payment, billing or logistics for the sale of crops.
- A separate mobile application, although Salesforce Mobile can open the same data.

1.6 Organisation of the Report

The remaining chapters of this report are organised as follows.

- **Chapter 2 - System Analysis:** studies the existing system, the proposed system, feasibility and requirements.

- **Chapter 3 - System Design:** explains the data model, objects, security and user-interface design.
- **Chapter 4 - Implementation:** describes every configuration step with the team's own screenshots.
- **Chapter 5 - Testing:** lists the test cases and their results.
- **Chapter 6 - Results and Discussion:** compares the outcome with the objectives.
- **Chapter 7 - Conclusion:** summarises the work and suggests future enhancements.

CHAPTER 2: SYSTEM ANALYSIS

2.1 Existing System

In the existing manual system, a farm manager writes the details of farmers, farms, crops and buyers in registers or in separate spreadsheet files. Contact numbers, locations and harvest dates are typed again whenever they are needed, and the connection between a crop and its farm exists only in the memory of the people who maintain the records.

2.2 Limitations of the Existing System

- Duplicate and inconsistent data, because the same detail is stored in several places.
- No relationship between records, so related information cannot be traced quickly.
- No access control; a spreadsheet is either fully shared or not shared at all.
- Manual calculation of yield figures, which can lead to mistakes.
- No standard way to produce reports or visual summaries.
- Risk of data loss if a register or a file is damaged or misplaced.

2.3 Proposed System

The proposed system stores all the information in Salesforce. Each kind of information has its own object, and the objects are related to each other. The Farm object belongs to a Farmer through a Master-Detail relationship, and the Crop object points to its Farm through a Lookup relationship. Buyers are stored in their own object so that sales contacts are kept together.

A Lightning app named Farm Management System gives users a single place to work. Access is controlled by the Farm Manager profile and role together with sharing rules, and the Farmer With Farms report and dashboard give an instant summary.

2.4 Advantages of the Proposed System

- **Single source of truth:** every record is stored once and reused through relationships.
- **Data quality:** picklists and the State-District dependency reduce spelling and selection errors.
- **Automatic calculation:** the formula field returns the expected yield per acre without manual work.
- **Security:** profile, role and sharing rules decide who can read or edit a record.
- **Insight:** reports and dashboards can be refreshed at any time.
- **Low cost and low maintenance:** no code and no server; an administrator can change the configuration.

2.5 Feasibility Study

Technical feasibility. All the required features are available as standard configuration in a Salesforce Developer Edition organisation. The team needed only a web browser and an internet connection, and the skills required were covered in the Naan Mudhalvan course.

Economic feasibility. The Developer Edition organisation is free for learning and development. No software licences, servers or other equipment had to be purchased, so the project has virtually no cost.

Operational feasibility. The Lightning interface is simple and consistent. Farm staff only need to open the app, choose a tab and fill in a form, so very little training is needed.

2.6 Software Requirements

Component	Specification
Platform	Salesforce Lightning Experience
Organisation type	Salesforce Developer Edition
Configuration tools	Setup, Object Manager, App Manager, Lightning App Builder, Report Builder, Dashboard Builder
Web browser	Google Chrome (latest version)
Operating system	Windows 10 or 11 (any OS with a modern browser works)
Programming language	None required (declarative configuration)

2.7 Hardware Requirements

Component	Minimum specification
Processor	Dual-core processor, 2 GHz or higher
Memory	4 GB RAM
Storage	1 GB free space
Network	Stable broadband or mobile data connection
Display	1366 x 768 resolution or higher

2.8 Functional Requirements

ID	Requirement
FR-01	The system shall store farmer details: name, contact number, state, district and village.
FR-02	The system shall store farm details: name, type, soil quality, irrigation type, location, size and the owning farmer.
FR-03	The system shall store crop details: name, farm, planting date, harvest date, actual yield and acres harvested.
FR-04	The system shall calculate the expected yield per acre automatically.

ID	Requirement
FR-05	The system shall store buyer details: name, type, contact, e-mail and address.
FR-06	The system shall show only the districts that belong to the selected state.
FR-07	The system shall restrict record access using a profile, a role and sharing rules.
FR-08	The system shall provide a report and a dashboard on farmers and their farms.

CHAPTER 3: SYSTEM DESIGN

3.1 System Architecture

The application is organised in layers. The user works at the top layer through the Lightning app, and every layer below it is configured in Salesforce Setup.

Layer	Components
Presentation	Farm Management System Lightning app, custom tabs (Farms, Crops, Farmers, Buyers), Reports and Dashboards tabs
Business logic	Formula field (Expected Yield per Acre), dependent picklist (State to District), relationships
Data	Custom objects Farm, Crop, Farmer and Buyer with their fields
Security	Farm Manager profile, Farm Manager role, Crop and Buyer sharing rules
Analytics	Report "Farmer With Farms" and dashboard "Farmers With Farms"
Platform	Salesforce Lightning Platform, Developer Edition organisation

3.2 Data Model

Figure 3.1 shows the entity-relationship view of the system. A farmer owns many farms and a farm grows many crops. The Buyer object stands alone and is shared with the sales team through a sharing rule.

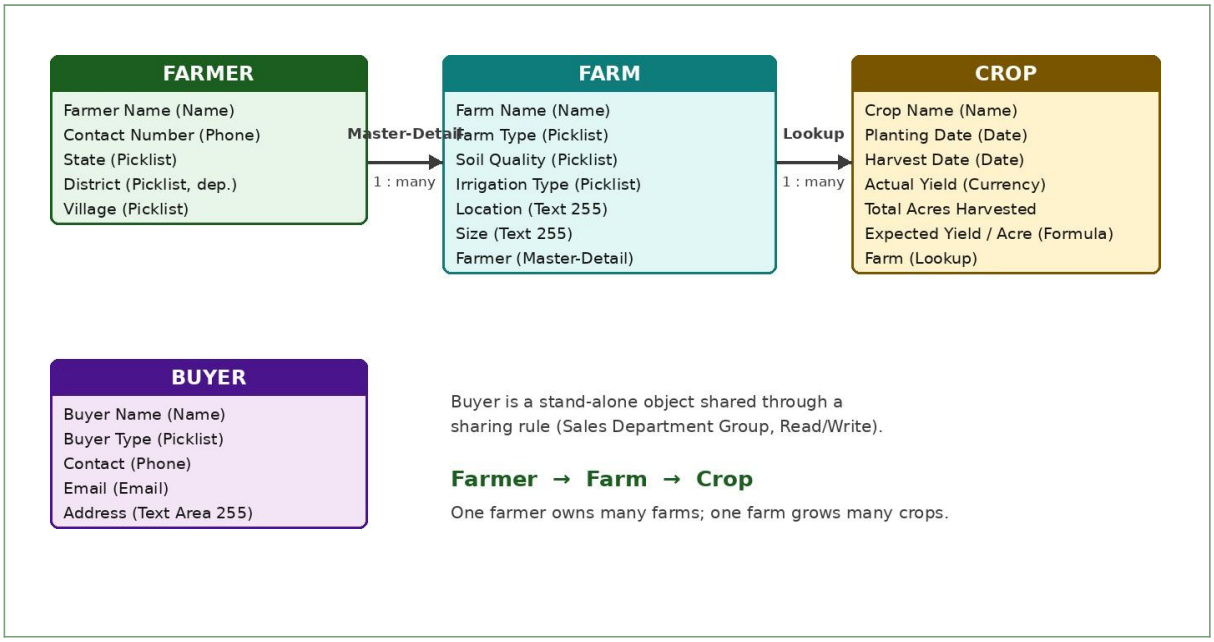


Figure 3.1: Entity-relationship diagram of the Farm Management System

Parent object	Child object	Relationship type
Farmer	Farm	Master-Detail: a farm cannot exist without its farmer and is deleted if the farmer is deleted

Parent object	Child object	Relationship type
Farm	Crop	Lookup: a crop points to the farm where it is grown

3.3 Object Design

The four custom objects and their fields, as configured in Object Manager, are listed below. Every object also carries the standard fields Created By, Last Modified By and, where shown, Owner.

3.3.1 Farm object

Field label	API name	Data type
Farm Name	Name	Text(80)
Farm Type	Farm_Type__c	Picklist
Farmer	Farmer__c	Master-Detail(Farmer)
Irrigation Type	Irrigation_Type__c	Picklist
Location	Location__c	Text(255)
Size	Size__c	Text(255)
Soil Quality	Soil_Quality__c	Picklist

3.3.2 Crop object

Field label	API name	Data type
Crop Name	Name	Text(80)
Crop Name	Crop_Name__c	Text(20)
Farm	Farm__c	Lookup(Farm)
Planting Date	Planting_Date__c	Date
Harvest Date	Harvest_Date__c	Date
Actual Yield	Actual_Yield__c	Currency(18, 0)
Total Acres Harvested	Total_Acres_Harvested__c	Number(18, 0)
Expected Yield per Acre	Expected_Yield_per_Acre__c	Formula (Currency)

3.3.3 Farmer object

Field label	API name	Data type
Farmer Name	Name	Text(80)
Contact Number	Contact_Number__c	Phone
State	State__c	Picklist (controlling)
District	District__c	Picklist (dependent on State)
Village	Village__c	Picklist

3.3.4 Buyer object

Field label	API name	Data type
Buyer Name	Name	Text(80)
Buyer Type	Buyer_Type__c	Picklist
Contact	Contact__c	Phone
Email	Email__c	Email
Address	Address__c	Text Area(255)

3.4 Dependent Picklist Design

On the Farmer object the State field controls the District field. When a user selects a state, the District picklist offers only the districts that were allowed for that state. This avoids impossible combinations such as a district that does not belong to the selected state, and it keeps the data consistent for reporting.

3.5 Security Design

Component	Design
Profile	Farm Manager - a custom profile based on the Salesforce user licence
Role	Farm Manager - added to the organisation role hierarchy
Crop sharing rule	Criteria-based rule that shares Crop records with the role Farm Manager at Read/Write access
Buyer sharing rule	Owner-based rule that shares Buyer records owned by the Sales Department Group with the same group at Read/Write access
Overrides	System Administrator and the Analytics Cloud Integration User profile can view all Crop records regardless of sharing rules

3.6 User Interface Design

The Farm Management System Lightning app uses the standard Lightning navigation bar with six items in this order: Farms, Crops, Farmers, Buyers, Reports and Dashboards. The app uses the primary colour #0070D2 and is shown in the App Launcher with the initials FM. Every object tab uses the Airplane tab style.

3.7 Report and Dashboard Design

The report "Farmer With Farms" uses the report type Farmers with Farms. It groups the data by farmer name and lists the farm name under each farmer, with row counts, subtotals and a grand total. The dashboard "Farmers With Farms" holds a gauge component whose source is this report.

CHAPTER 4: IMPLEMENTATION

4.1 Implementation Overview

This chapter describes how the Farm Management System was built, step by step. Each of the following sections explains one configuration activity and shows the corresponding screenshot taken from the team's own Salesforce organisation. The activities were carried out in the order shown in the table below.

Step	Activity	Setup path	Figure
1	Create the Farm, Crop, Farmer and Buyer objects with fields	Object Manager	4.2 - 4.5
2	Create the State-District dependency	Farmer > Field Dependencies	4.6
3	Create the object tabs	Tabs	4.7 - 4.8
4	Create the Lightning app	App Manager	4.9 - 4.11
5	Create the profile, role and review users	Profiles, Roles, Users	4.12 - 4.14
6	Configure sharing rules	Sharing Settings	4.15 - 4.16
7	Enter sample records	Farm Management System app	4.17 - 4.21
8	Create the report and dashboard	Reports, Dashboards	4.22 - 4.23
9	Open the app from the App Launcher	App Launcher	4.24

Screenshots in this chapter were captured in Google Chrome while logged in as the team leader, L.M Sriraksha.

4.2 Farm Object - Fields and Relationships

The Farm object is the central record of the system. It stores the physical details of each farm and is tied to its owner through a Master-Detail relationship with the Farmer object.

Steps performed

1. Open Setup, choose Object Manager and select the Farm object.
2. Under Fields & Relationships create Farm Type, Soil Quality and Irrigation Type as picklists.
3. Create Location and Size as Text(255) fields.
4. Create the Farmer field as a Master-Detail(Farmer) relationship.

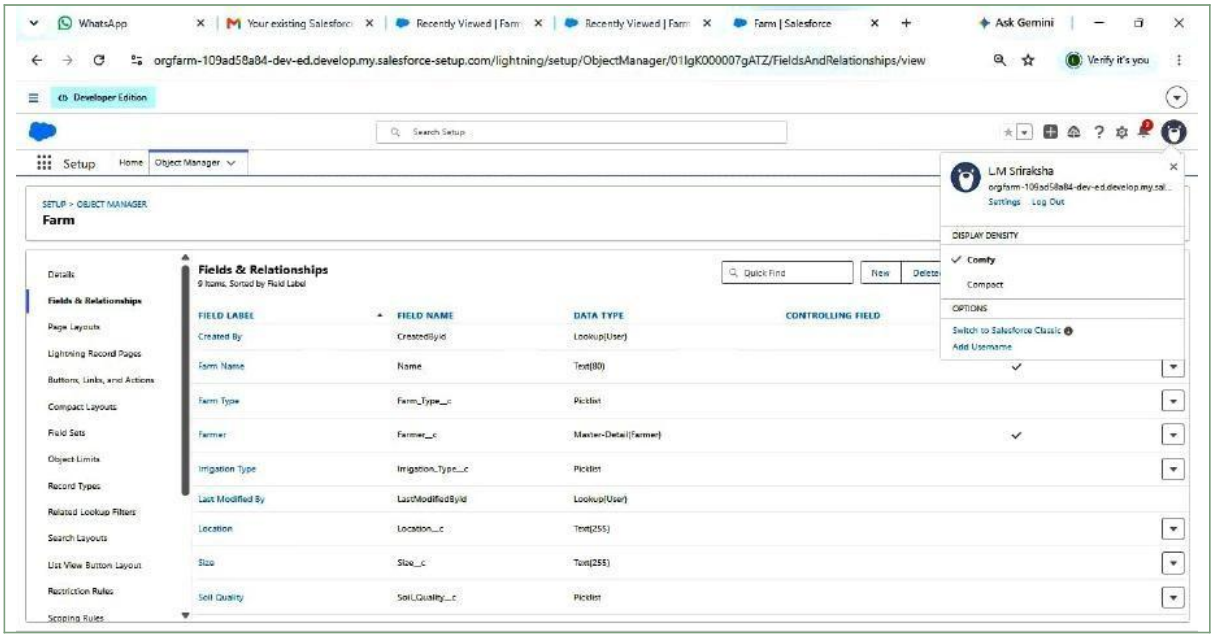


Figure 4.2: Farm object: Fields & Relationships page

Observation: The page lists 9 items: the standard fields Created By and Last Modified By, the Name field labelled Farm Name (Text 80) and six custom fields, with Farmer shown as Master-Detail(Farmer).

4.3 Crop Object - Fields and Relationships

The Crop object records what is grown on a farm and how much it yields. It also contains the formula field that calculates the expected yield per acre.

Steps performed

1. Open Object Manager and select the Crop object.
2. Create Farm as a Lookup(Farm) field.
3. Create Planting Date and Harvest Date as Date fields, Actual Yield as Currency(18, 0) and Total Acres Harvested as Number(18, 0).
4. Create Expected Yield per Acre as a formula field with the return type Currency.

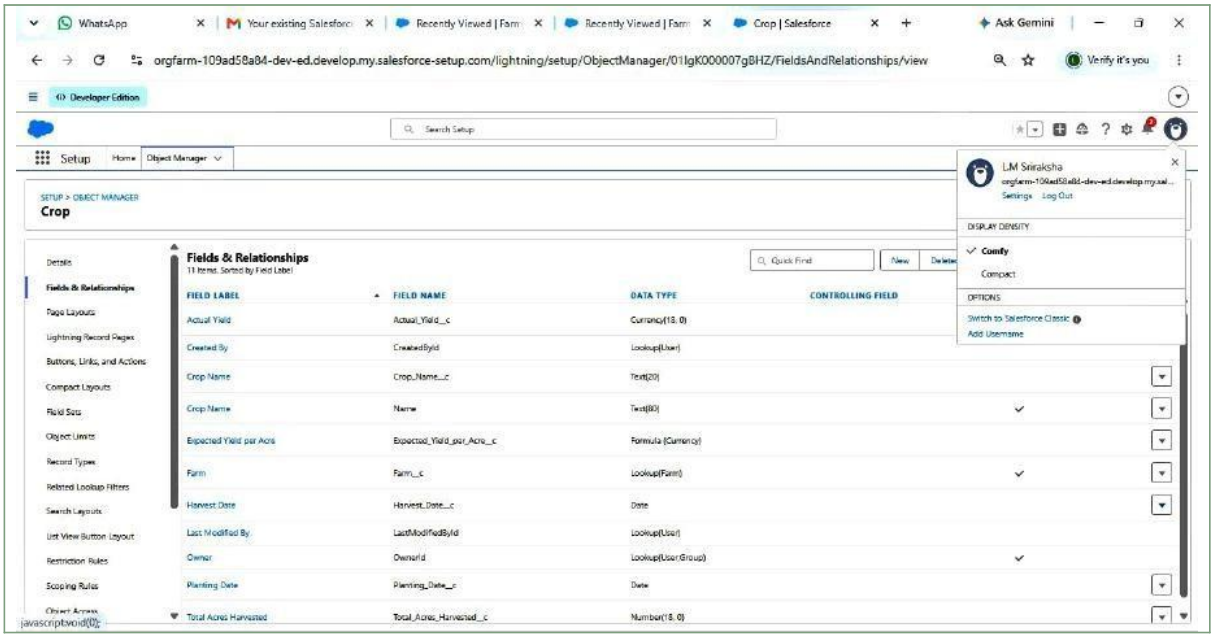


Figure 4.3: Crop object: Fields & Relationships page

Observation: The page lists 11 items. The Rice record later returns $2,000.00 for an actual yield of $10,000 over 5 acres, which shows that the formula divides Actual Yield by Total Acres Harvested.

4.4 Farmer Object - Fields and Relationships

The Farmer object holds the personal and location details of each farmer. It is the parent object in the Master-Detail relationship with Farm.

Steps performed

1. Open Object Manager and select the Farmer object.
2. Create Contact Number as a Phone field.
3. Create State, District and Village as picklist fields.
4. Make State the controlling field of District (see Section 4.6).

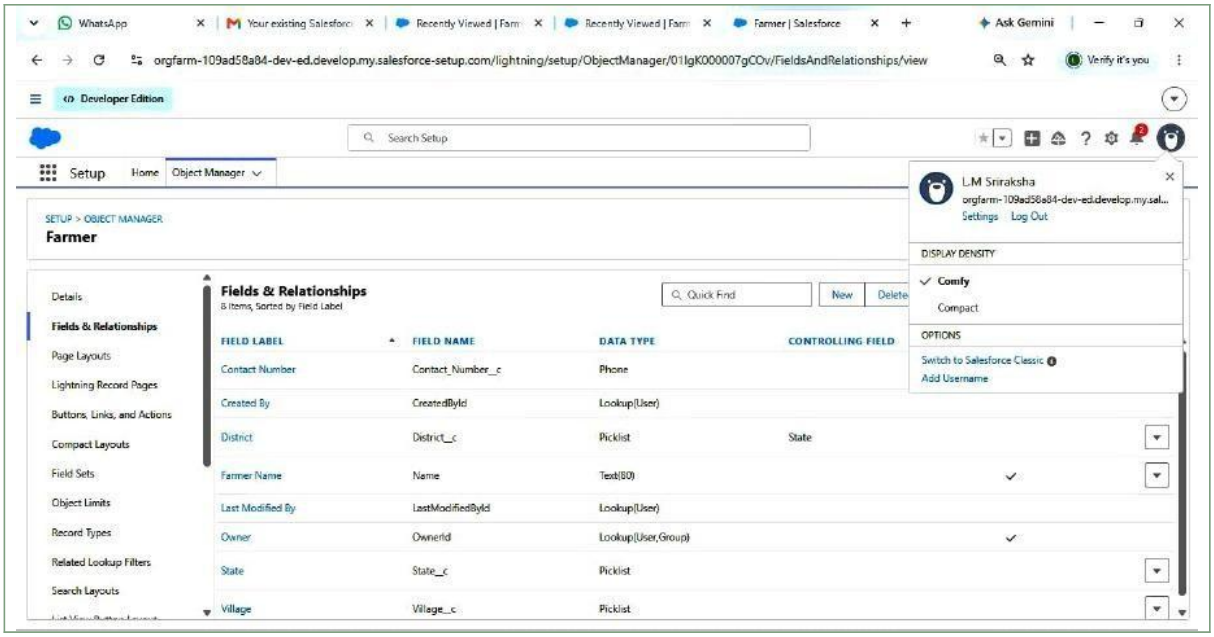


Figure 4.4: Farmer object: Fields & Relationships page

Observation: The page lists 8 items. The District row shows State in the Controlling Field column, confirming the dependency.

4.5 Buyer Object - Fields and Relationships

The Buyer object stores the people and firms that purchase the harvested crops, together with their contact details.

Steps performed

1. Open Object Manager and select the Buyer object.
2. Create Buyer Type as a picklist.
3. Create Contact as a Phone field and Email as an Email field.
4. Create Address as a Text Area(255) field.

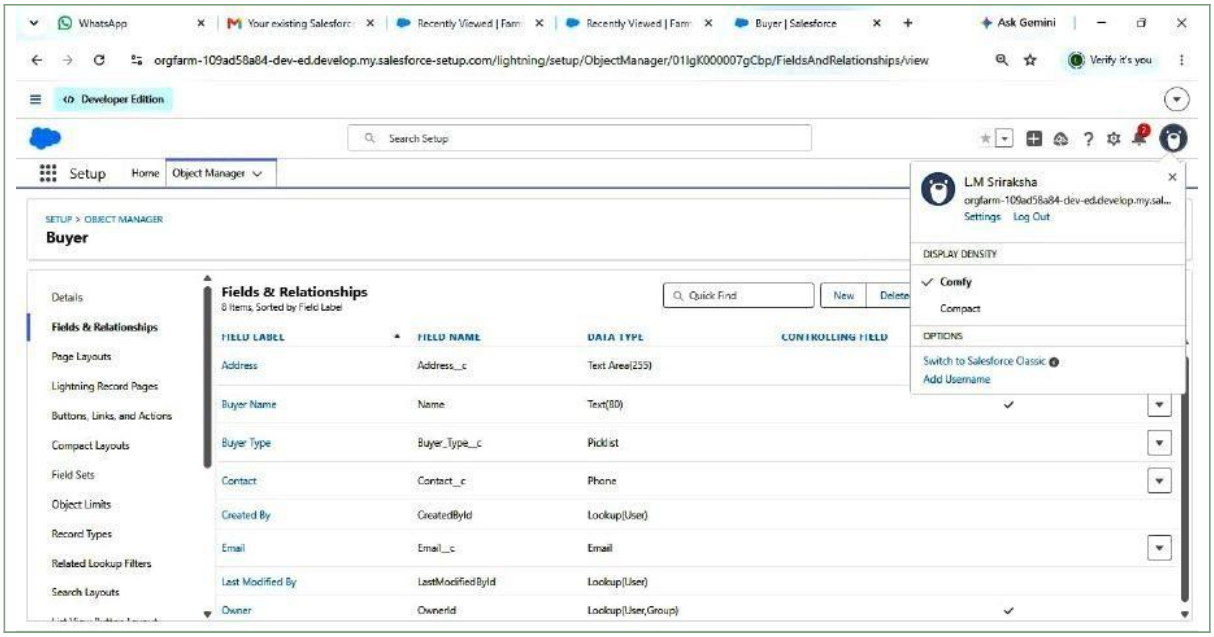


Figure 4.5: Buyer object: Fields & Relationships page

Observation: The page lists 8 items: Address, Buyer Name, Buyer Type, Contact, Email and the standard fields Created By, Last Modified By and Owner.

4.6 Dependent Picklist - State and District

A field dependency makes the values of one picklist depend on the value chosen in another. Here the district choices depend on the selected state.

Steps performed

1. Open the Farmer object and go to Fields & Relationships.
2. Click Field Dependencies and then New.
3. Select State as the controlling field and District as the dependent field.
4. Choose the districts allowed for each state and save.

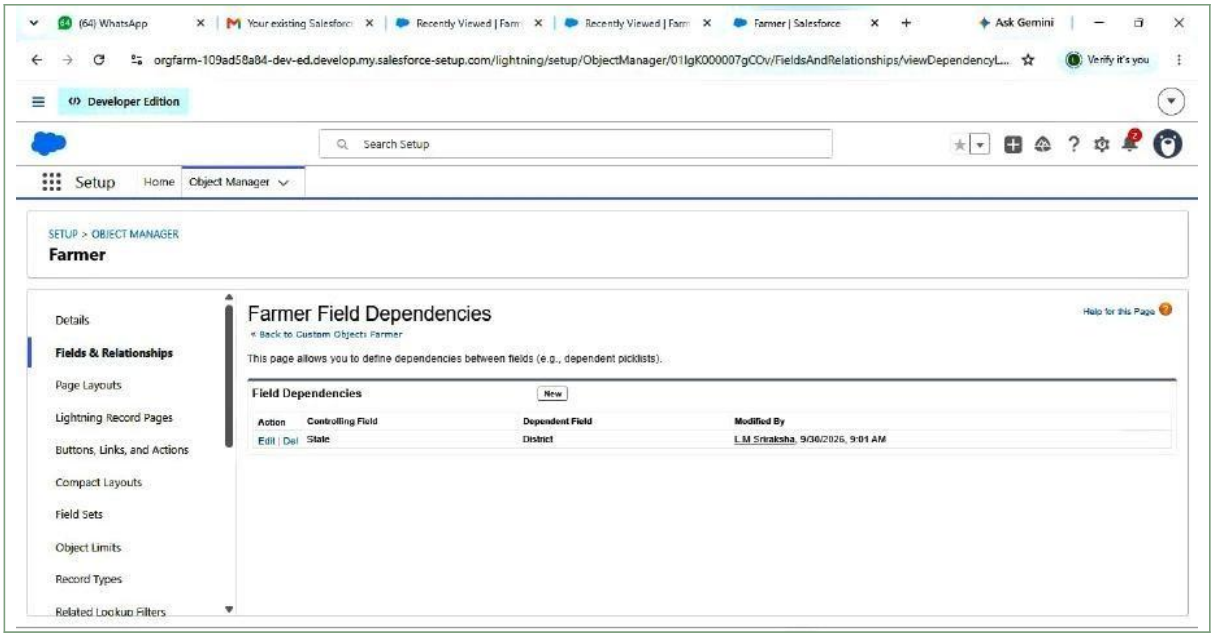


Figure 4.6: Farmer Field Dependencies page

Observation: The Field Dependencies list shows one entry with State as the controlling field and District as the dependent field, modified by L.M Sriraksha on 9/30/2026.

4.7 Custom Object Tabs

A tab makes an object available in the navigation bar. A custom tab was created for each of the four objects.

Steps performed

1. In Setup search for Tabs and open the Tabs page.
2. In the Custom Object Tabs section click New.
3. Select the object, choose the tab style Airplane and continue.
4. Make the tab visible to the required profiles and save.

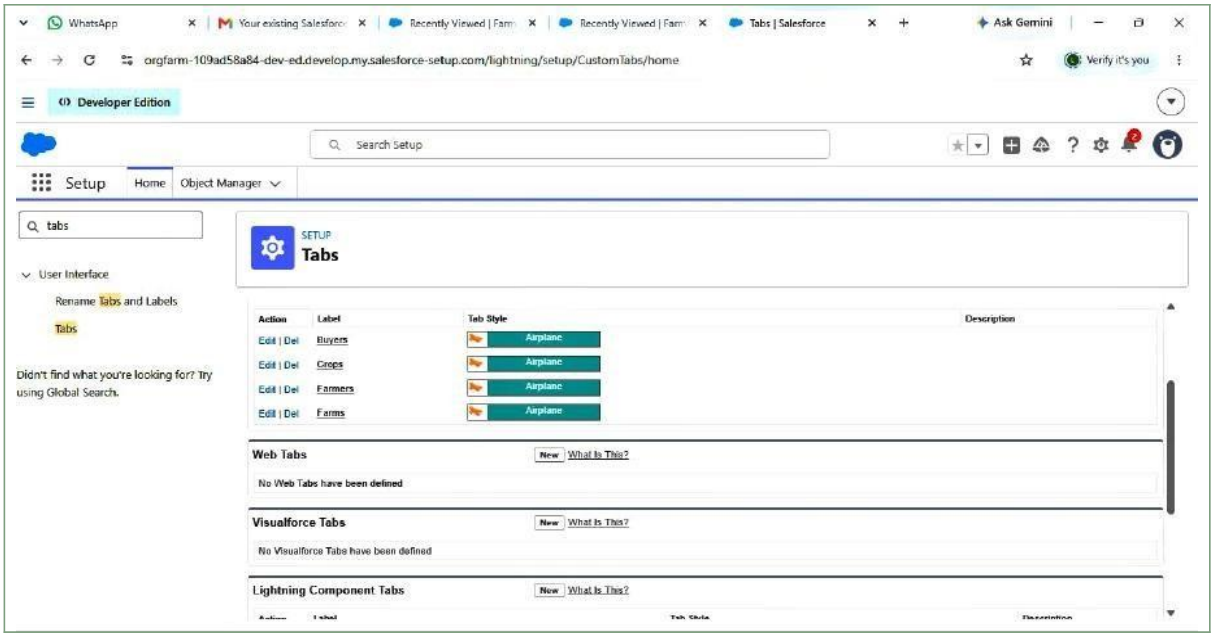


Figure 4.7: Setup > Tabs: custom object tabs

Observation: The Custom Object Tabs list shows Buyers, Crops, Farmers and Farms, each with the Airplane tab style.

4.8 Lightning Component Tabs

Lower on the same Tabs page, Salesforce lists the Web Tabs, Visualforce Tabs, Lightning Component Tabs and Lightning Page Tabs of the organisation.

Steps performed

1. Scroll down on the Tabs page.
2. Check the Lightning Component Tabs section.
3. Check the Lightning Page Tabs section.

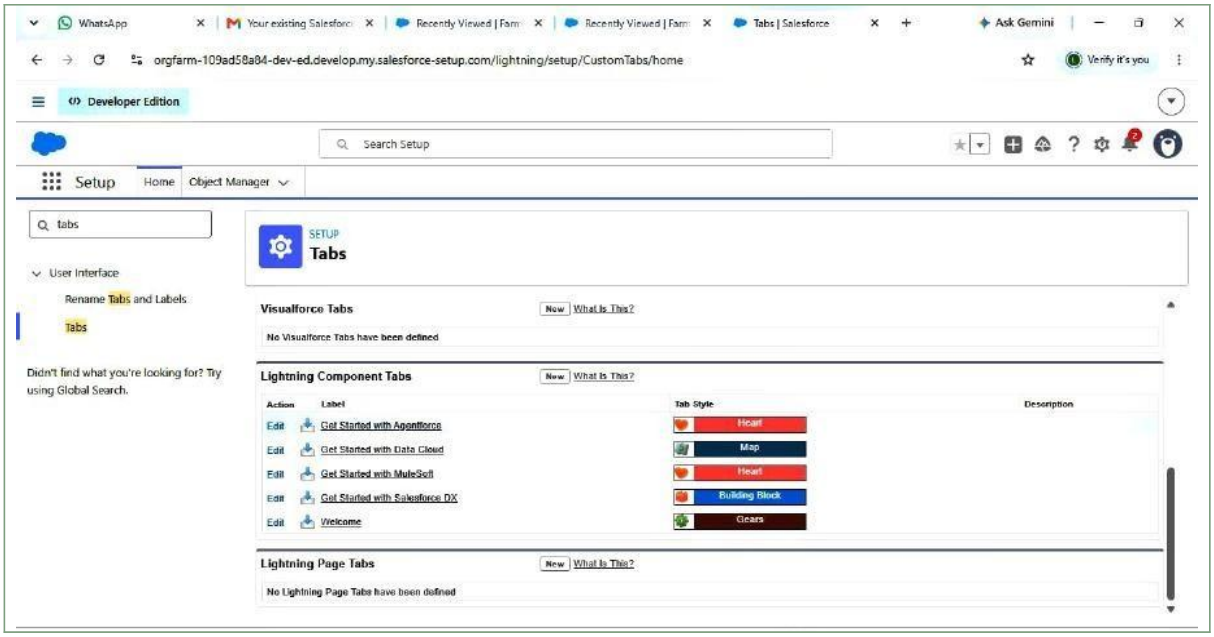


Figure 4.8: Setup > Tabs: Lightning component and page tabs

Observation: The Lightning Component Tabs section shows the default tabs of a Developer Edition organisation (Get Started with Agentforce, Data Cloud, MuleSoft and Salesforce DX, and Welcome). They were not created by the team. No Lightning Page Tabs have been defined.

4.9 Lightning Experience App Manager

The App Manager lists every app in the organisation and is the starting point for creating a new Lightning app.

Steps performed

1. In Setup search for App Manager and open it.
2. Click New Lightning App.
3. Enter the app details and navigation items (next two sections).
4. Save; the app then appears in the list.

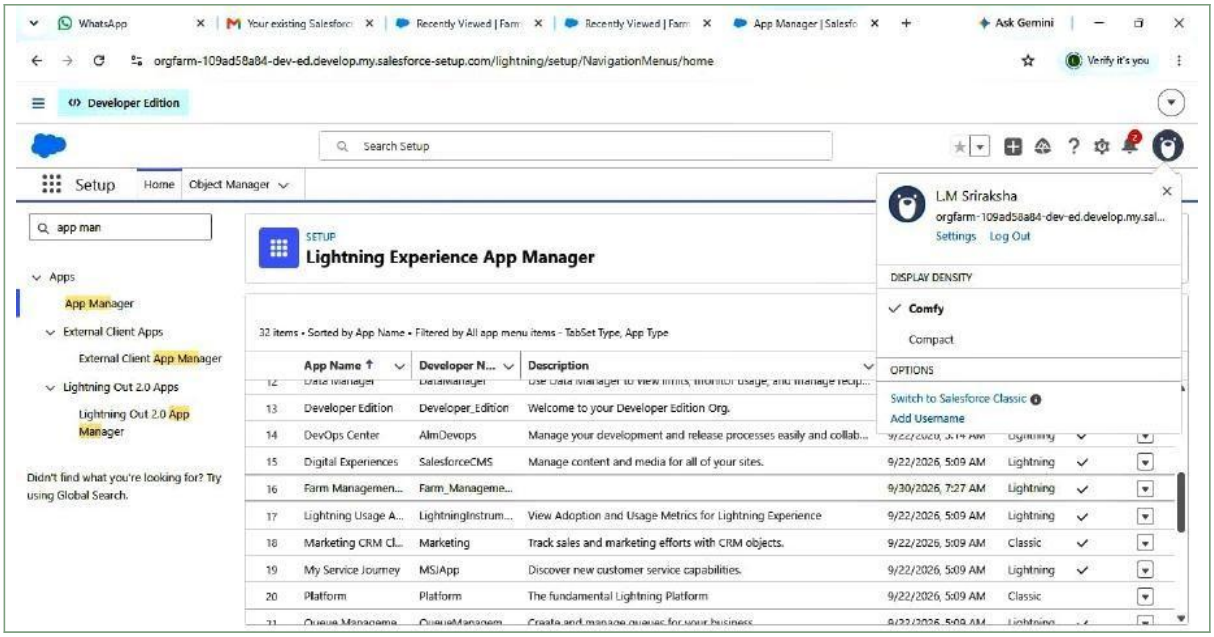


Figure 4.9: Lightning Experience App Manager

Observation: The list shows 32 items. Row 16 is the Farm Management System app (developer name Farm_Management_System) of type Lightning, last modified on 9/30/2026.

4.10 App Details and Branding

This page gives the app its name, developer name and look. The branding colour is used in the navigation bar and in the App Launcher.

Steps performed

1. Enter Farm Management System as the App Name.
2. Enter Farm_Management_System as the Developer Name.
3. Keep the primary colour #0070D2.
4. Check the App Launcher Preview and click Next.

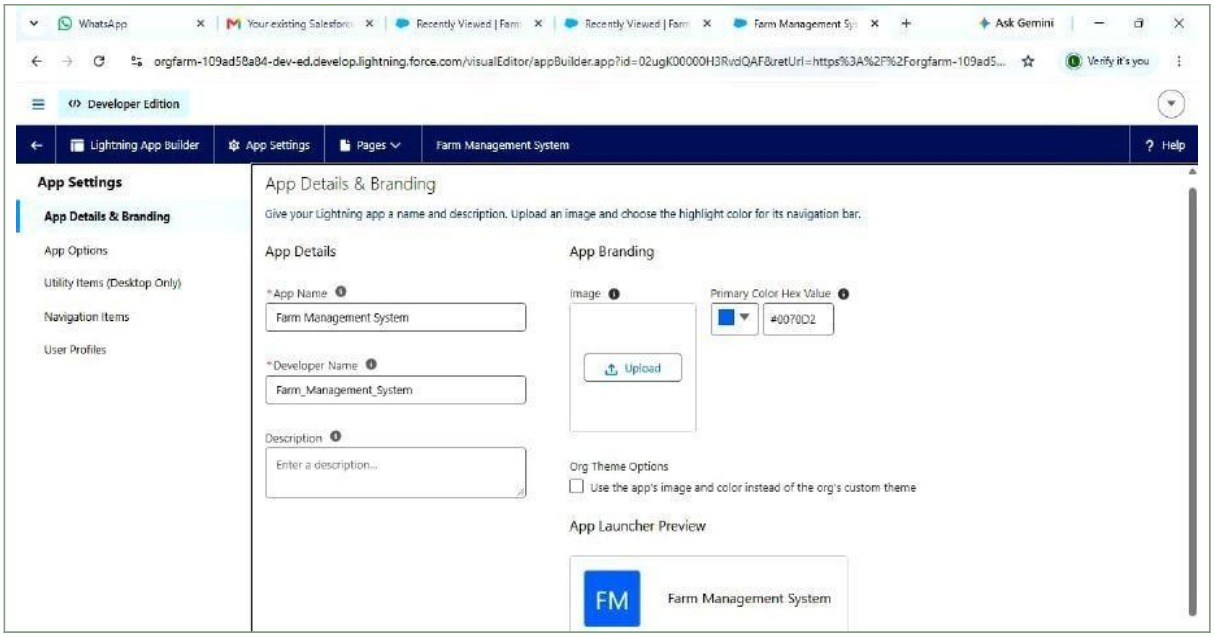


Figure 4.10: Lightning App Builder: App Details & Branding

Observation: The App Launcher Preview shows a blue tile with the initials FM next to the name Farm Management System.

4.11 App Navigation Items

Navigation items decide which tabs appear in the app and in which order.

Steps performed

1. Open the Navigation Items page of the app settings.
2. Move Farms, Crops, Farmers, Buyers, Reports and Dashboards from Available Items to Selected Items.
3. Arrange the order with the up and down arrows.
4. Assign user profiles and save.

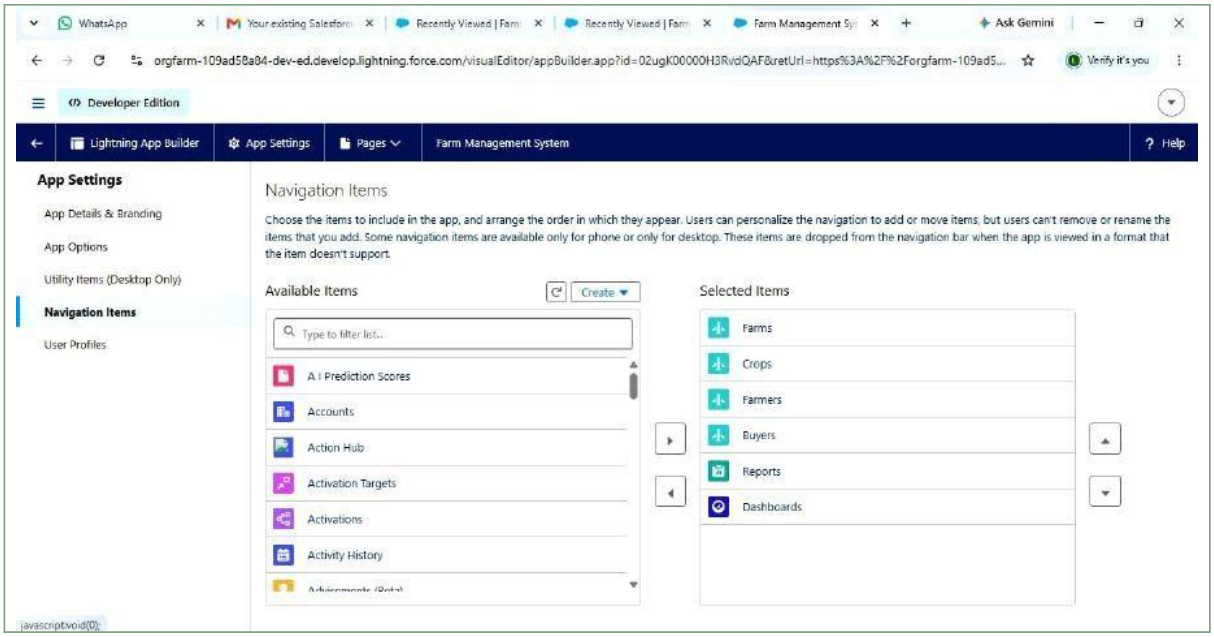


Figure 4.11: Lightning App Builder: Navigation Items

Observation: The Selected Items list shows the six items in the required order: Farms, Crops, Farmers, Buyers, Reports and Dashboards.

4.12 Profiles

A profile defines what a user can do with each object. A custom profile named Farm Manager was created for the staff who run the farms.

Steps performed

1. In Setup search for Profiles.
2. Click New Profile, or clone an existing profile.
3. Name the profile Farm Manager and select the Salesforce user licence.
4. Set the object permissions and save.

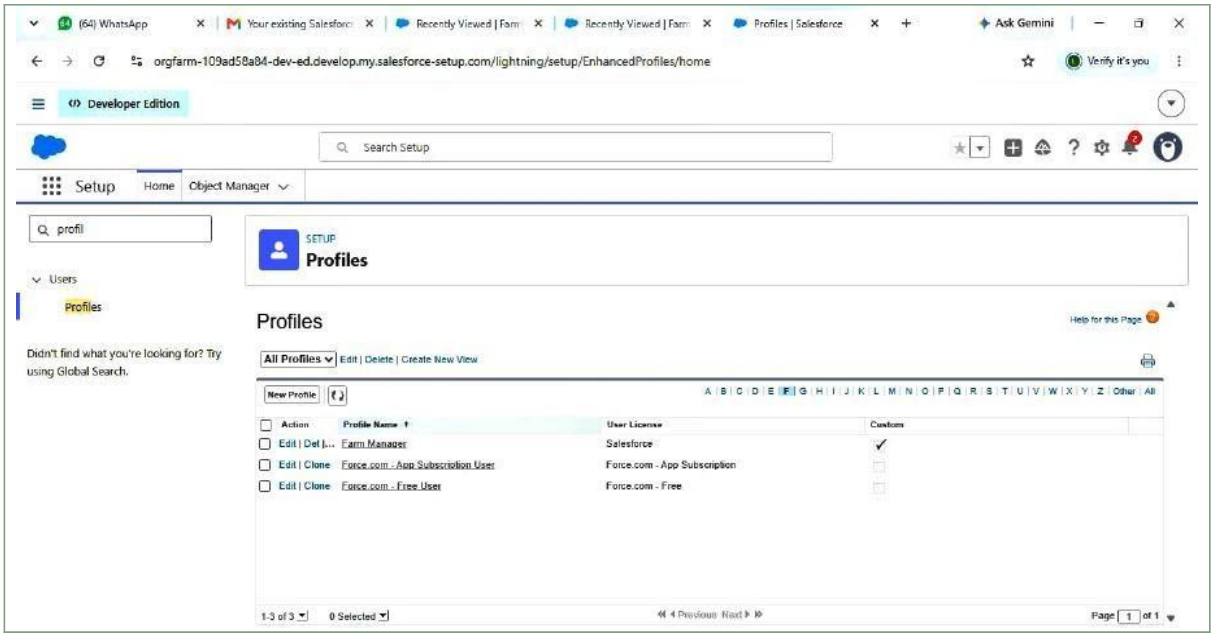


Figure 4.12: Setup > Profiles: Farm Manager profile

Observation: The list (filter F) shows three profiles: Farm Manager (custom, Salesforce licence), Force.com - App Subscription User and Force.com - Free User.

4.13 Roles and Role Hierarchy

Roles control record visibility up and down the organisation hierarchy. A Farm Manager role was added to the hierarchy.

Steps performed

1. In Setup search for Roles.
2. Click Add Role at the required position.
3. Enter Farm Manager as the role name and save.
4. Use Assign to attach users to the role.

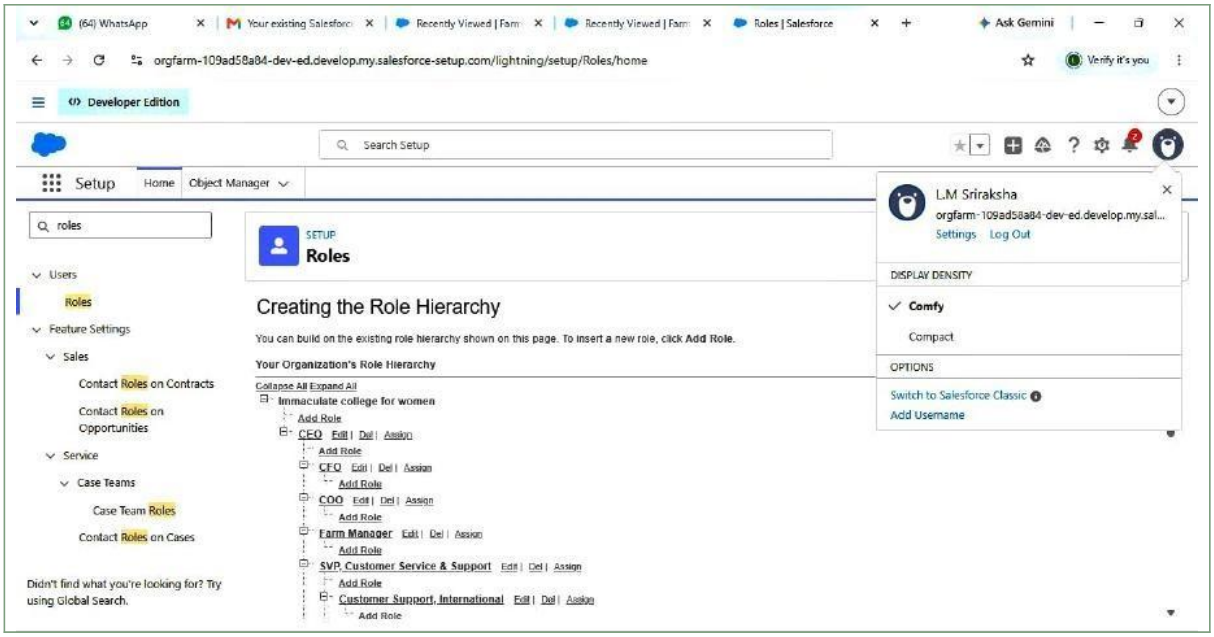


Figure 4.13: Setup > Roles: role hierarchy

Observation: The hierarchy begins with the organisation name Immaculate college for women. The Farm Manager role appears alongside CFO and COO under the CEO, with SVP, Customer Service & Support also listed.

4.14 Users

The Users page lists the people who can log in to the organisation. Each user has a profile and, optionally, a role.

Steps performed

1. In Setup search for Users.
2. Choose the view All Users.
3. Use the alphabet bar to filter the list.
4. Click New User to add a user and select the profile and role.

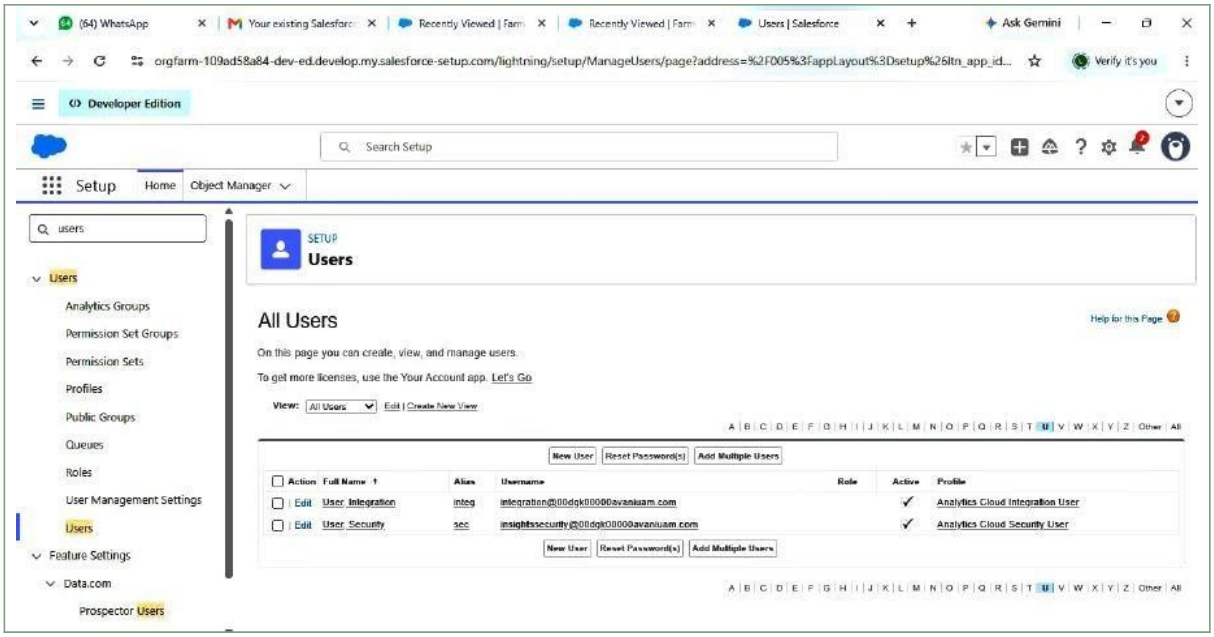


Figure 4.14: Setup > Users: All Users list

Observation: With the letter U selected, the list shows two active Analytics Cloud system users: User, Integration and User, Security.

4.15 Sharing Rule for the Crop Object

Sharing rules open up record access beyond the default. A rule was created to share Crop records with the Farm Manager role.

Steps performed

1. In Setup open Sharing Settings.
2. Under Crop Sharing Rules click New.
3. Define the criteria and select Role: Farm Manager as the group to share with.
4. Set the access level to Read/Write and save.

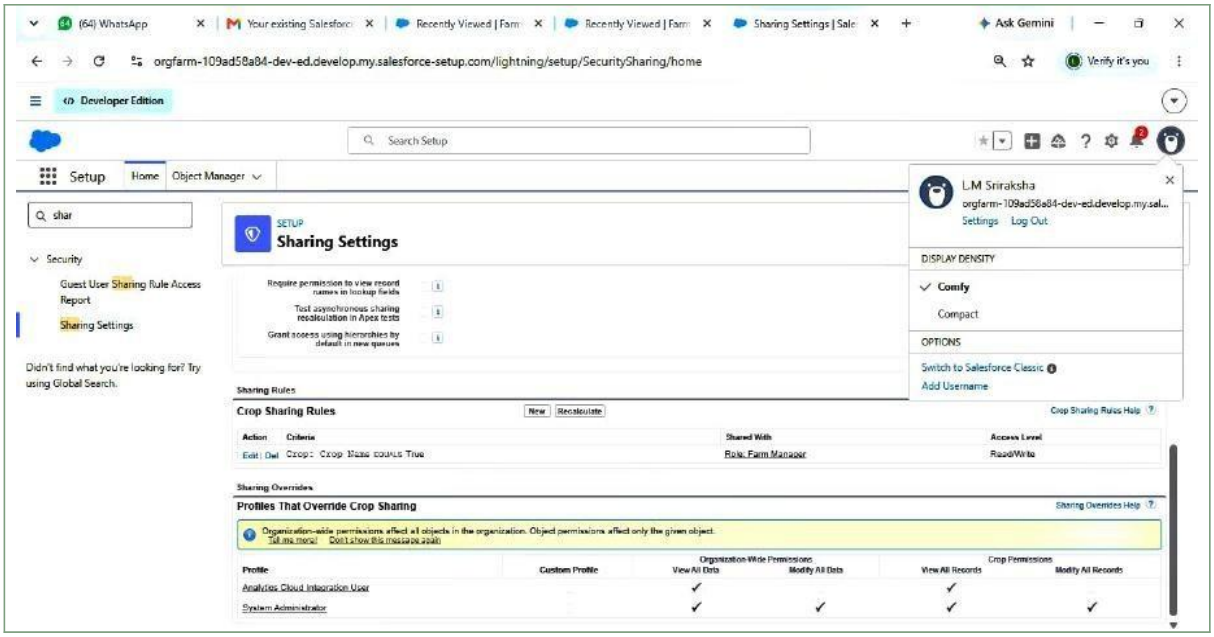


Figure 4.15: Setup > Sharing Settings: Crop Sharing Rules

Observation: The Crop Sharing Rules table shows the rule shared with Role: Farm Manager at Read/Write access. The overrides table shows that System Administrator has View All and Modify All rights.

4.16 Sharing Rule for the Buyer Object

A second sharing rule makes Buyer records available to the sales team.

Steps performed

1. On the Sharing Settings page scroll to Buyer Sharing Rules.
2. Click New and select an owner-based rule.
3. Choose Group: Sales Department Group as the owner group and as the group to share with.
4. Set the access level to Read/Write and save.

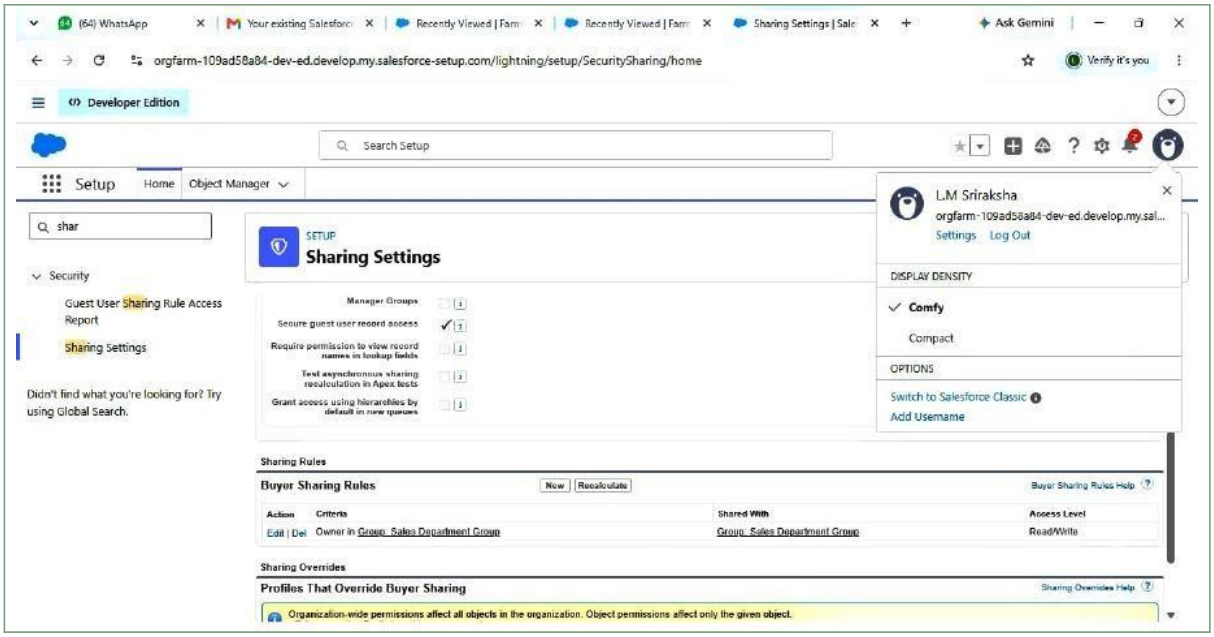


Figure 4.16: Setup > Sharing Settings: Buyer Sharing Rules

Observation: The Buyer Sharing Rules table shows Owner in Group: Sales Department Group, shared with the same group at Read/Write access.

4.17 Farms List View

After the app was ready, records were created from its tabs. The Farms tab opens the list view of recently viewed farms.

Steps performed

1. Open the Farm Management System app from the App Launcher.
2. Click the Farms tab.
3. Select the Recently Viewed list view.

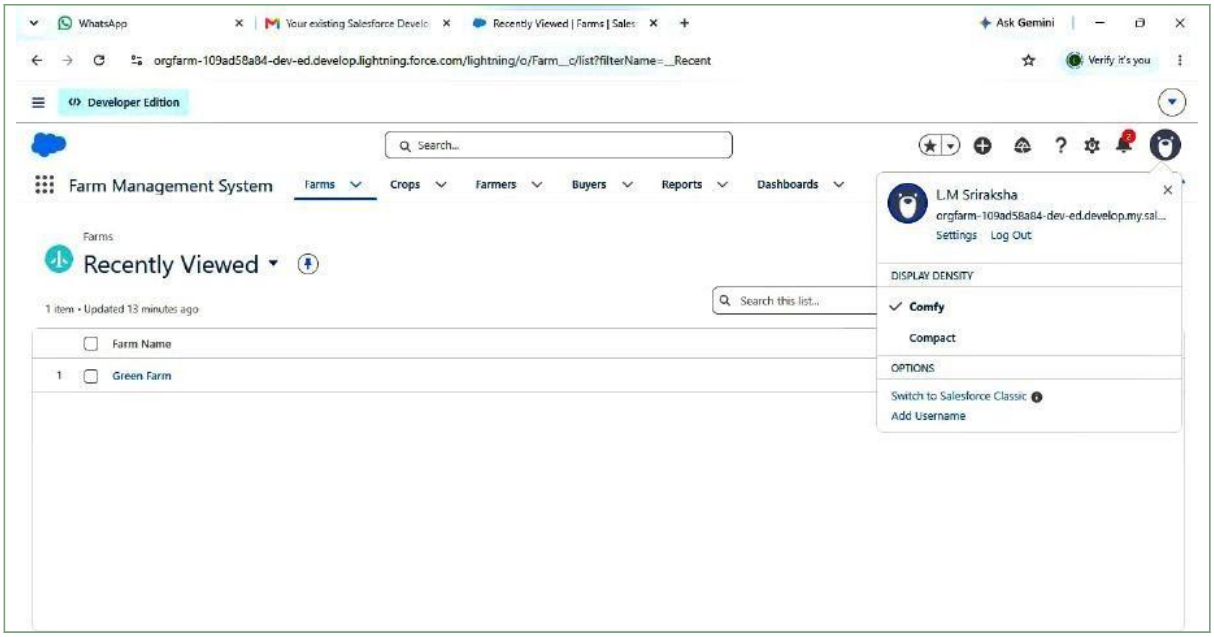


Figure 4.17: Farm Management System app: Farms list view

Observation: The list shows 1 item, Green Farm. The navigation bar shows the six tabs of the app.

4.18 Farm Record - Green Farm

The first sample farm links the farmer to the land on which the crop is grown.

Steps performed

1. On the Farms tab click New.
2. Enter the farm name, type, soil quality, irrigation type, location and size.
3. Select the farmer through the Farmer lookup and save.

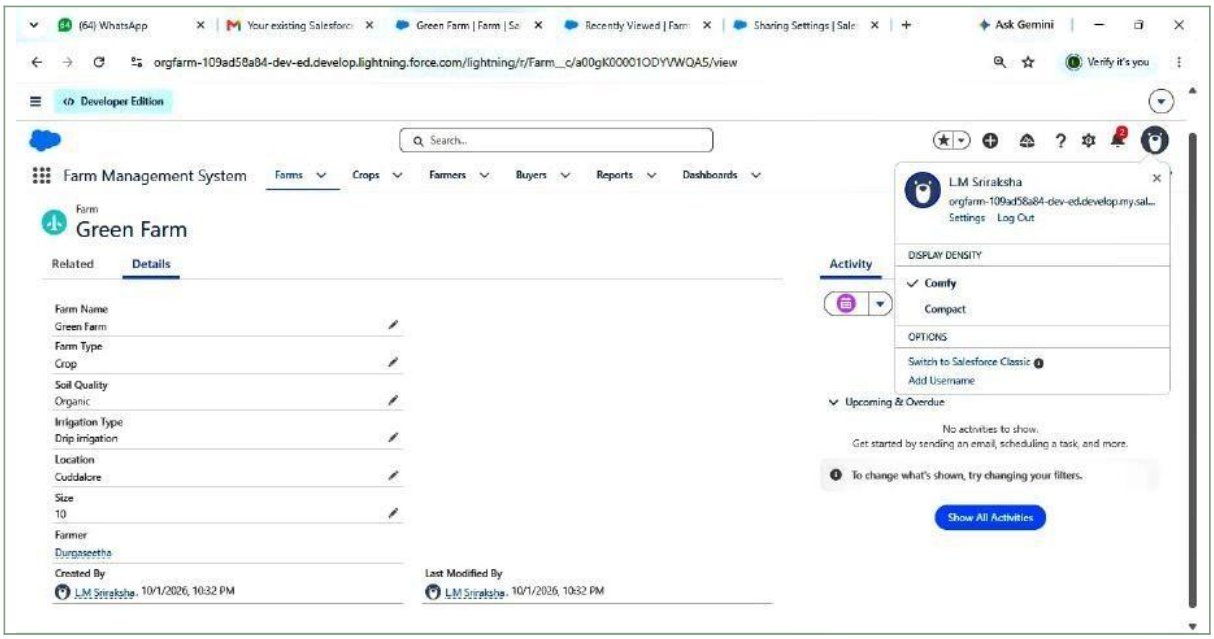


Figure 4.18: Farm record: Green Farm

Observation: Green Farm has Farm Type Crop, Soil Quality Organic, Irrigation Type Drip irrigation, Location Cuddalore and Size 10, and belongs to the farmer Durgaseetha.

4.19 Crop Record - Rice

The sample crop shows the Lookup relationship to the farm and the formula field in action.

Steps performed

1. On the Crops tab click New.
2. Enter the crop name, planting date, harvest date, actual yield and total acres harvested.
3. Select Green Farm in the Farm lookup and save.

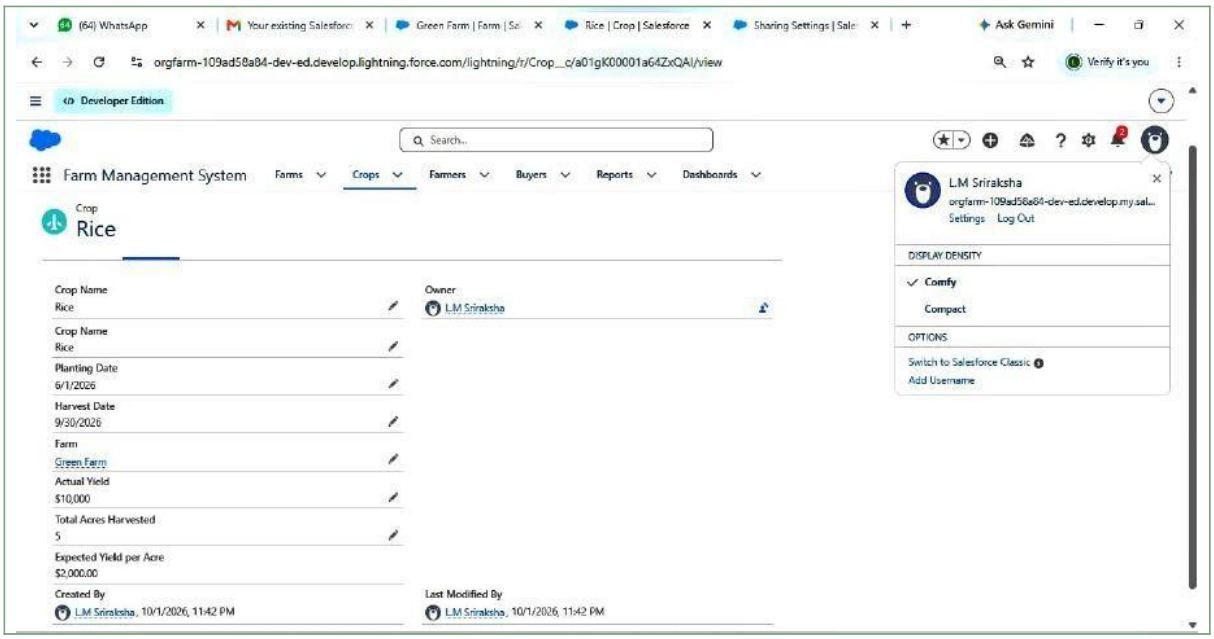


Figure 4.19: Crop record: Rice

Observation: Rice was planted on 6/1/2026 and harvested on 9/30/2026 on Green Farm, with an actual yield of $10,000 over 5 acres. The formula field returns an Expected Yield per Acre of $2,000.00.

4.20 Farmer Record - Durgaseetha

The sample farmer record demonstrates the Phone field and the State-District dependency.

Steps performed

1. On the Farmers tab click New.
2. Enter the farmer name and contact number.
3. Select the State first and then the District from the filtered list.
4. Select the village and save.

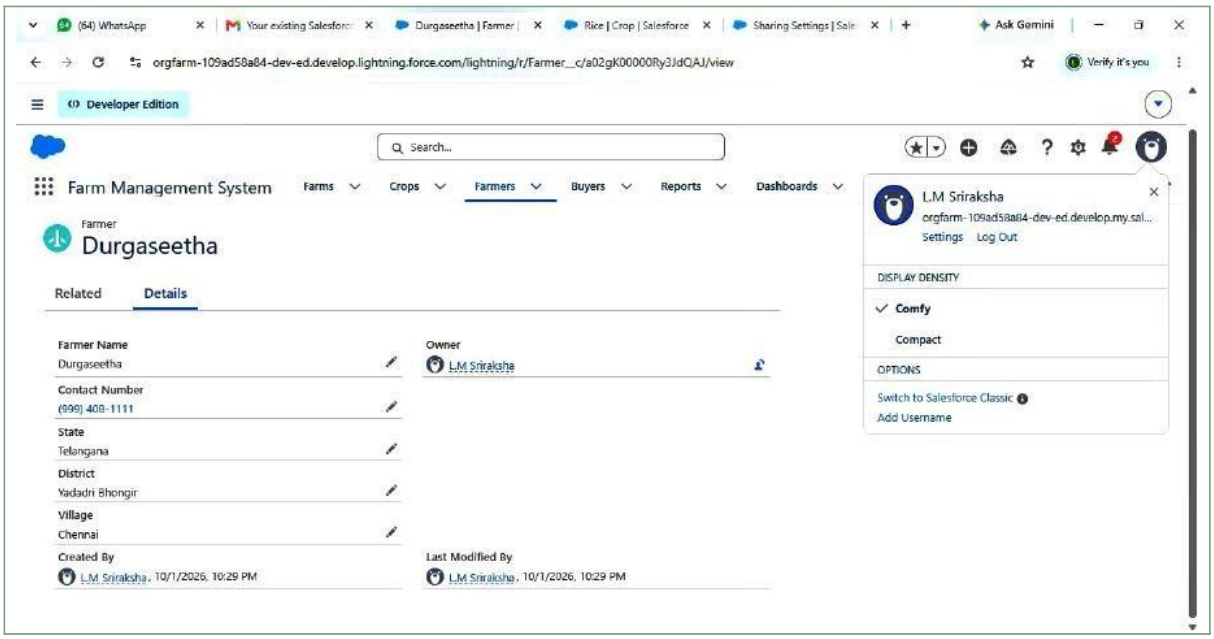


Figure 4.20: Farmer record: Durgaseetha

Observation: Durgaseetha has the contact number (999) 408-1111, State Telangana, District Yadadri Bhongir and Village Chennai.

4.21 Buyer Record - LMS Traders

The sample buyer shows the Phone, Email, Text Area and picklist fields of the Buyer object.

Steps performed

1. On the Buyers tab click New.
2. Enter the buyer name, contact number, e-mail address and address.
3. Choose the buyer type and save.

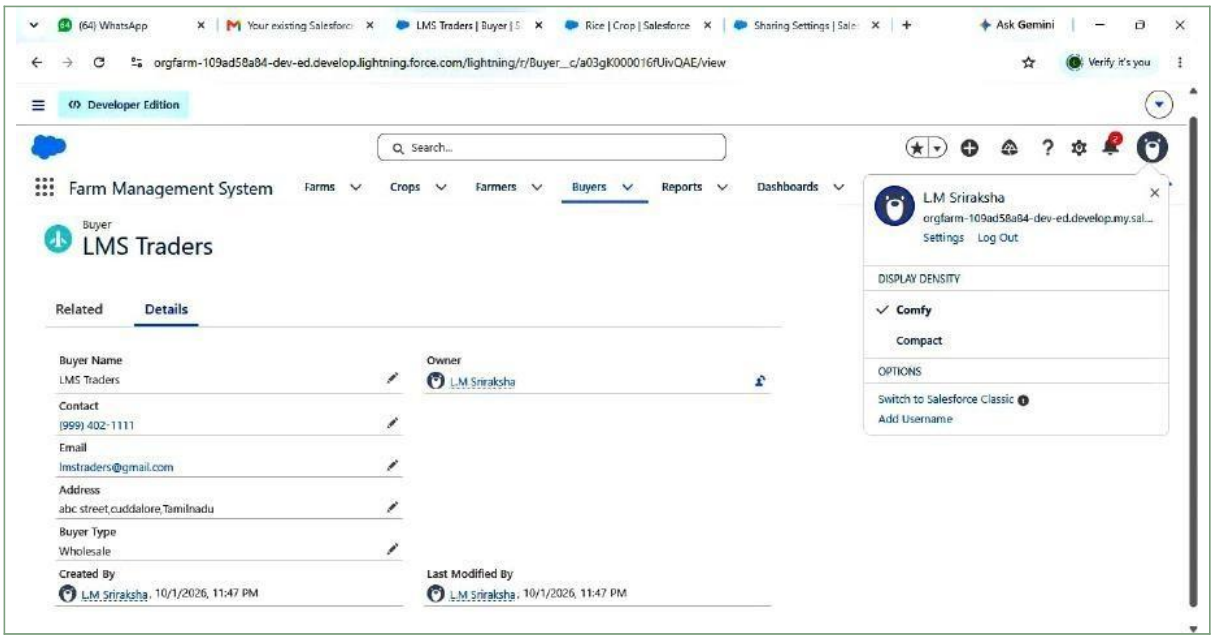


Figure 4.21: Buyer record: LMS Traders

Observation: LMS Traders is a Wholesale buyer with the contact number (999) 402-1111, the e-mail address lmstraders@gmail.com and the address abc street, cuddalore, Tamilnadu.

4.22 Report - Farmer With Farms

The report lists every farmer with the farms that belong to him. It uses the report type Farmers with Farms, which follows the Master-Detail relationship.

Steps performed

1. Open the Reports tab and click New Report.
2. Select the report type Farmers with Farms and start the report.
3. Group the rows by Farmer Name and show the Farm Name column.
4. Save the report as Farmer With Farms.

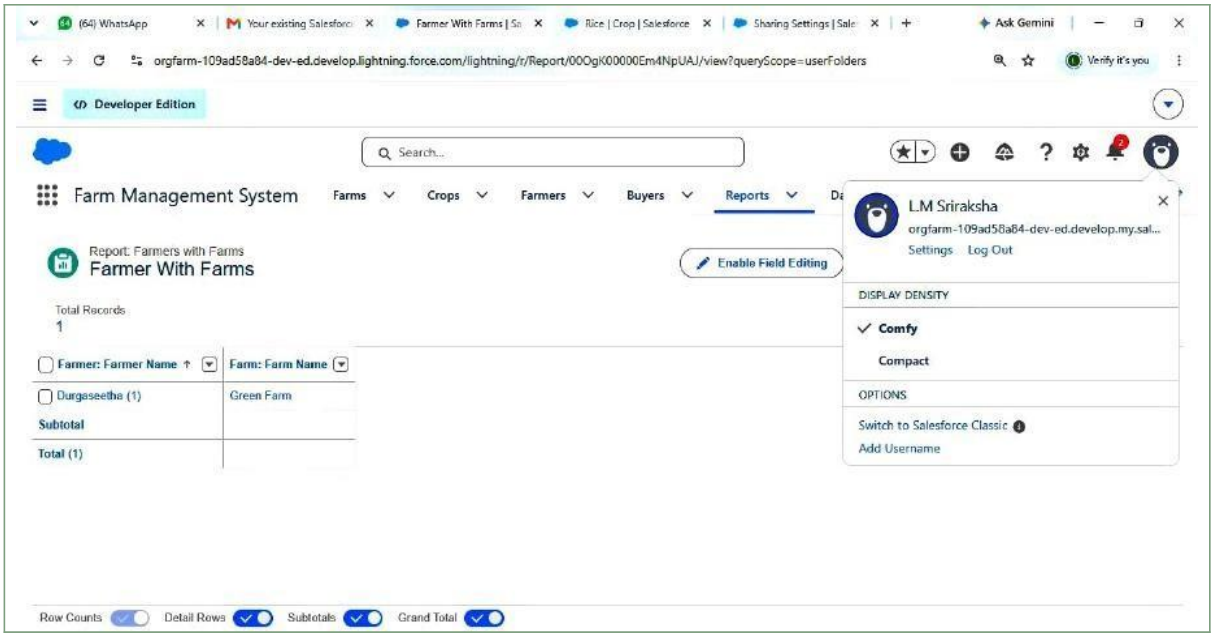


Figure 4.22: Report: Farmer With Farms

Observation: The report shows Total Records 1: the farmer Durgaseetha (1) with the farm Green Farm. Row Counts, Detail Rows, Subtotals and Grand Total are switched on.

4.23 Dashboard - Farmers With Farms

The dashboard turns the report into a visual component that can be read at a glance.

Steps performed

1. Open the Dashboards tab and click New Dashboard.
2. Add a Gauge component.
3. Select the Farmer With Farms report as the source.
4. Save the dashboard as Farmers With Farms.

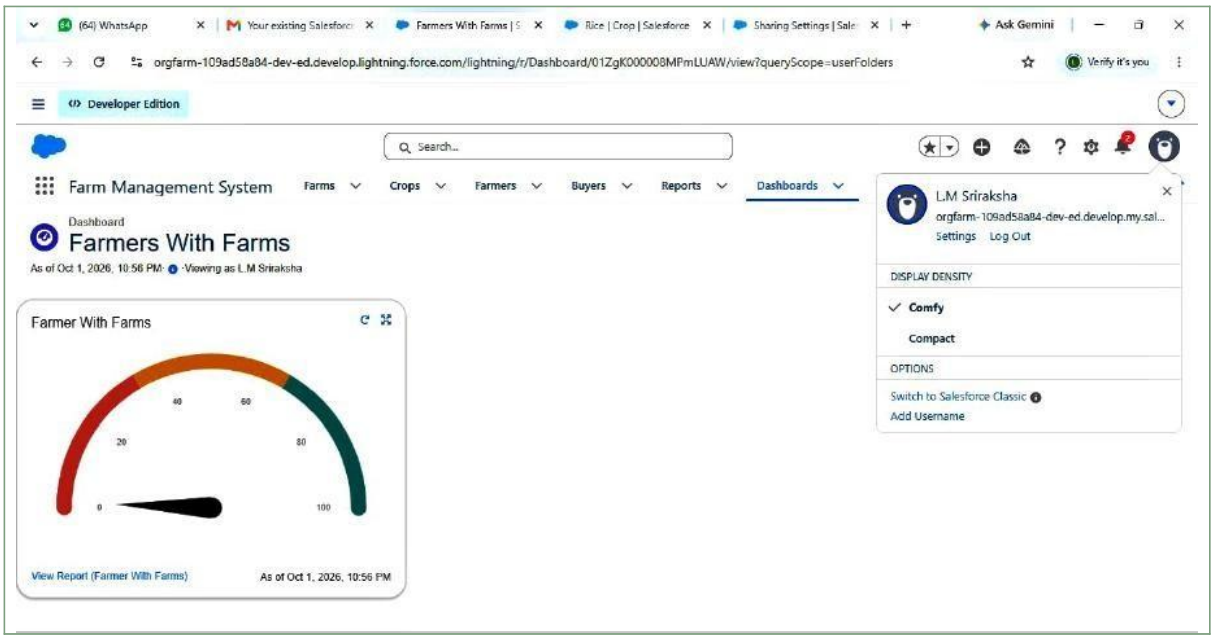


Figure 4.23: Dashboard: Farmers With Farms

Observation: The dashboard, viewed as L.M Sriraksha on Oct 1, 2026, shows a gauge scaled from 0 to 100 with a link to View Report (Farmer With Farms).

4.24 App Launcher

The App Launcher is where users find all the apps available to them, including the new application.

Steps performed

1. Click the nine-dot App Launcher icon at the top left.
2. Scroll to the list of apps.
3. Click Farm Management System to open it.

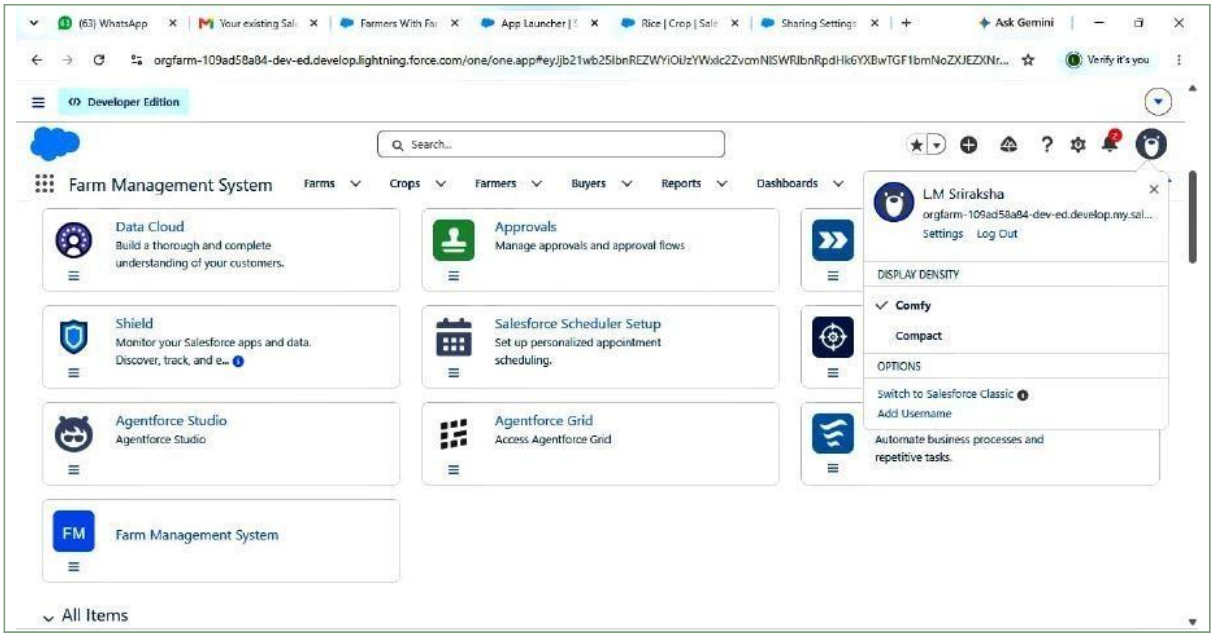


Figure 4.24: App Launcher showing Farm Management System

Observation: The Farm Management System app, with the FM icon, appears in the App Launcher alongside the standard Salesforce apps such as Data Cloud, Approvals and Agentforce Studio.

CHAPTER 5: TESTING

5.1 Testing Objectives

Testing was carried out to confirm that every object, relationship, rule and screen behaves as designed. Because the application is built by configuration, the tests concentrate on whether the data is stored correctly, whether the relationships and the formula behave as expected, and whether the security components are in place.

5.2 Test Environment

Item	Detail
Organisation	Salesforce Developer Edition
Interface	Lightning Experience
Browser	Google Chrome
Tested by	Project team
Test data	Durgaseetha, Green Farm, Rice and LMS Traders

5.3 Functional Test Cases

TC	Test case	Expected result	Result
01	Create a Farmer record	Record saves with name, contact, state, district and village	Pass (Fig 4.20)
02	Select State, then District	District list is limited to the chosen state	Pass (Fig 4.6)
03	Create a Farm linked to a Farmer	Farm saves; Farmer appears as a link	Pass (Fig 4.18)
04	Create a Crop linked to a Farm	Crop saves; Farm appears as a link	Pass (Fig 4.19)
05	Check Expected Yield per Acre	$10,000 over 5 acres gives $2,000.00	Pass (Fig 4.19)
06	Create a Buyer record	Phone, e-mail, address and type are stored	Pass (Fig 4.21)
07	Open the Farms list view	Created farm is listed	Pass (Fig 4.17)
08	Open every tab of the app	Farms, Crops, Farmers, Buyers, Reports and Dashboards open	Pass (Fig 4.11)
09	Run the report	Shows Durgaseetha with Green Farm, total 1	Pass (Fig 4.22)
10	Open the dashboard	Gauge component loads from the report	Pass (Fig 4.23)

TC	Test case	Expected result	Result
11	Find the app in the App Launcher	Farm Management System is listed	Pass (Fig 4.24)

5.4 Security Configuration Checks

TC	Check	Expected result	Result
S1	Farm Manager profile exists	Custom profile with Salesforce licence is listed	Pass (Fig 4.12)
S2	Farm Manager role exists	Role appears in the hierarchy	Pass (Fig 4.13)
S3	Crop sharing rule	Shared with Role: Farm Manager, Read/Write	Pass (Fig 4.15)
S4	Buyer sharing rule	Shared with Sales Department Group, Read/Write	Pass (Fig 4.16)

5.5 Test Summary

Category	Test cases	Passed
Functional	11	11
Security configuration	4	4
Total	15	15

ADVANTAGES AND DISADVANTAGES

Advantages

- No programming is needed; the whole application is built with standard Salesforce configuration.
- Farm, crop, farmer and buyer records are stored once and connected through relationships.
- The State-District dependency reduces selection and spelling errors during data entry.
- The formula field calculates the expected yield per acre automatically.
- The Farm Manager profile, role and sharing rules control who can view and edit each record.
- The report and dashboard give an instant summary of farmers and their farms.
- Being cloud-based, the system needs no server and can be opened from any web browser.
- The Developer Edition organisation is free, and an administrator can change the configuration without code.

Disadvantages

- The application needs an internet connection to be used.
- A Developer Edition organisation is meant for learning and development, not for live production use.
- Only one sample record exists for each object, so the report and dashboard show very small figures.
- Farm Size is a text field, so sizes cannot be added up in reports.
- There is no Sale object yet, so the sale of a crop to a buyer is not recorded.
- The Crop sharing rule is based on the Crop Name field and must be widened to cover all crops.
- Without Apex or Flow there is no automation, such as reminders before the harvest date.

CHAPTER 6: RESULTS AND DISCUSSION

6.1 Results Achieved

Objective	Outcome	Evidence
Separate objects for farms, crops, farmers, buyers	Four custom objects created with their fields	Figs 4.2 - 4.5
Related records	Master-Detail (Farm to Farmer) and Lookup (Crop to Farm) working	Figs 4.18, 4.19
Cleaner data entry	District list depends on the selected State	Fig 4.6
Automatic calculation	Expected Yield per Acre returns $2,000.00 for the sample crop	Fig 4.19
Access control	Farm Manager profile and role, two sharing rules	Figs 4.12 - 4.16
One application	Lightning app with six navigation items	Figs 4.9 - 4.11
Reporting	Report and dashboard created	Figs 4.22, 4.23

6.2 Discussion

The project shows that a complete, usable business application can be assembled from standard Salesforce features alone. The relationships keep the records consistent: the report returns the farm of each farmer without any manual matching, and the formula field removes the need to calculate the yield per acre by hand.

At the time of the screenshots the dashboard gauge rests at the 0 end of its 0 to 100 scale. This is most likely because the organisation holds only one farmer and one farm, so the value shown by the report is very small compared with the scale. Adding more sample data or adjusting the gauge segment ranges will make the reading more informative.

6.3 Limitations

- Only one sample record exists for each object, so the report and dashboard show very small figures.
- The Crop sharing rule is criteria-based on the Crop Name field; it can be widened so that all crops are shared with the Farm Manager role.
- There is no Sale object yet, so the sale of a crop to a buyer is not recorded.
- Farm Size is a text field, so sizes cannot be added up in reports. A number field would allow totals.

CHAPTER 7: CONCLUSION AND FUTURE ENHANCEMENTS

7.1 Conclusion

The Farm Management System meets the objectives set at the start of the project. The team created four custom objects, connected them with Master-Detail and Lookup relationships, added a formula field and a dependent picklist, defined a profile, a role and sharing rules, and packaged everything in a Lightning app with a report and a dashboard.

The work gave the team practical experience of the Salesforce platform: data modelling in Object Manager, record security, app building and analytics. The same approach can be extended to larger agricultural organisations by adding more objects and automation.

7.2 Future Enhancements

- Add a Sale object to link crops to buyers, with quantity, price and date.
- Add validation rules to make sure the harvest date is not earlier than the planting date.
- Use Salesforce Flow to send reminders before the expected harvest date.
- Change the Farm Size field to a number so that total area can be reported.
- Create more dashboards, such as yield by crop and farms by district.
- Use record types to separate crop farms from other farm types.
- Use the Salesforce Mobile App so that field staff can update records on site.

7.3 Learning Outcomes

- Designing a data model with custom objects, fields and relationships.
- Using formula fields and dependent picklists to improve data quality.
- Applying profiles, roles and sharing rules to secure records.
- Building a Lightning app, a report and a dashboard.
- Working as a team on a cloud platform.

APPENDIX

A. Sample Data Used for Testing

Object	Record	Key details
Farmer	Durgaseetha	Contact (999) 408-1111; State Telangana; District Yadadri Bhongir; Village Chennai
Farm	Green Farm	Farm Type Crop; Soil Quality Organic; Irrigation Type Drip irrigation; Location Cuddalore; Size 10; Farmer Durgaseetha
Crop	Rice	Farm Green Farm; planted 6/1/2026; harvested 9/30/2026; Actual Yield $10,000; 5 acres harvested
Buyer	LMS Traders	Wholesale; contact (999) 402-1111; lmstraders@gmail.com; abc street, cuddalore, Tamilnadu

B. Formula Used

The Expected Yield per Acre field on the Crop object is a Currency formula field that divides Actual Yield by Total Acres Harvested (Actual_Yield__c / Total_Acres_Harvested__c). For the Rice record, $10,000 over 5 acres returns $2,000.00.

C. Glossary of Terms

- Custom object: a table created by the user to store one kind of record, such as Farm or Crop.
- Master-Detail relationship: a parent-child link in which the child record cannot exist without its parent.
- Lookup relationship: a link from one record to another that does not make the child depend on the parent.
- Dependent picklist: a picklist whose values depend on the value chosen in a controlling picklist.
- Formula field: a read-only field whose value is calculated automatically from other fields.
- Sharing rule: a rule that opens up record access to a role or group beyond the default.

REFERENCES

1. Salesforce Trailhead - https://trailhead.salesforce.com
2. Salesforce Help - https://help.salesforce.com
3. Salesforce Developer Documentation - https://developer.salesforce.com/docs
4. Naan Mudhalvan Portal - https://naanmudhalvan.tn.gov.in

THANK YOU

FARM MANAGEMENT SYSTEM

A Salesforce Lightning Application for Managing Farms, Crops, Farmers and Buyers